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Fuzzy Bayesian Knowledge Tracing

Fei Liu, Xuegang Hu*, Chenyang Bu*, and Kui Yu

***Abstract*—Online education promotes the sharing of learning resources. Knowledge tracing (KT) is aimed at tracking the cognition function of students according to their performance on various exercises at different times and has attracted considerable attention. Existing KT models primarily use bisection representations for the performance and cognitive states of students, thus limiting the application scope of these models and the accuracy of the evaluation of student cognitive performance in learning processes. Therefore, fuzzy Bayesian knowledge tracing models (namely, FBKT and T2FBKT) are proposed to address continuous score scenarios (e.g., subjective examinations) so that the applicability of KT models may be broadened. Moreover, fine-grained cognitive states can be discerned. In particular, referring to type-2 fuzzy theory, T2FBKT mitigates the model uncertainty of FBKT induced by uncertain parameters. Finally, extensive experiments demonstrate the effectiveness of the proposed fuzzy KT models.**

***Index Terms*—Educational data mining, knowledge tracing, fuzzy theory, hidden Markov model.**

I. Introduction

In the past decades, the development of online education has provided students with easy access to learning resources [1]. Knowledge tracing (KT) is aimed at modeling the process of mastering student knowledge over time based on the performance on various exercises [2]. As KT facilitates the evaluation of the learning effectiveness of students and the prediction future performance on exercises as students interact with online education systems, it is important in educational data mining [1] and knowledge engineering with big data [3]–[5].

Corbett et al. [6] first proposed a Bayesian KT (BKT) model based on the hidden Markov model (HMM) to describe the changing cognitive states of students during knowledge acquisition. In this model, the observation variable is the user performance score, and the hidden variable is the cognitive state of the user. Subsequently, many improvements have been proposed, including KT models based on the Bayesian method [7]–[9], deep learning methods [10]–[13], and various data attention mechanisms considering, for example, task-related information [14], [15], knowledge information [16]–[18], and time interval information [19], [20]). However, existing KT models primarily focus on bisection performance scores and bisection cognitive states [21], leading to the following two limitations.

First, bisection performance scores imply that regardless of whether a student provides an answer to a subjective or objective question, existing KT models can only handle 0/1 scores (correct/wrong). However, continuous scores are frequently used in many scenarios (e.g., scores on subjective questions and comprehensive scores on objective questions). For example, if a student does not obtain the correct answer at the first attempt but finally obtains it, or the student obtains the correct answer after some hints, the comprehensive score will not be 1 because of penalties related to multiple attempts and the provision of hints.

Second, bisection-cognitive-state KT models limit the evaluation accuracy of student cognitive states. Specifically, they model the evolution of student cognition using two states: "unknown" and "known" (i.e., 0 and 1). Continuous cognitive states can provide more accurate measures for students. For example, the bisection cognitive sequence $(0, 0, 0, 0, 1, 1, 1)$ denotes the "unknown" or "known" cognitive state of a student at each time step. Representing this sequence using continuous quantities, such as $(0.0, 0.2, 0.3, 0.2, 0.5, 0.7, 1.0)$, may indicate subtle changes in these cognitive states for educational resource recommendations based the individual needs of the student.

To summarize, the limitations of most existing KT models are as follows:

- They are not suitable for the continuous-score scenarios, such as subjective-question and credit scores.
- They model student cognition using only the "known" and "unknown" states and thus they ignore "partially known" in the learning process.

To address these limitations, we propose two fuzzy KT models (namely, FBKT and T2FBKT) as fuzzy theory can address concepts such as "partially correct" and "partially known" [22]. In particular, T2FBKT is based on the type-2 fuzzy theory and addresses model uncertainty (also known as epistemic uncertainty) in KT models. Model uncertainty is induced by uncertain parameters in the mapping between the input and the output of a model [23], and type-2 fuzzy theory has been proved successful in addressing this issue [23].

Specifically, the contributions of this study are as follows (the relationship between our motivations and contributions is summarized in Fig. 1):

- FBKT is proposed to address the limitations of bisection representations in existing KT models. Fuzzy scores, cognitive states, and their fuzzy relations assist KT in adapting to continuous score scenarios and capture fine-grained changes in student cognition.
- Furthermore, T2FBKT is proposed to mitigate model uncertainty in the FBKT model through the type-2 fuzzifications of the relations between fuzzy cognitive

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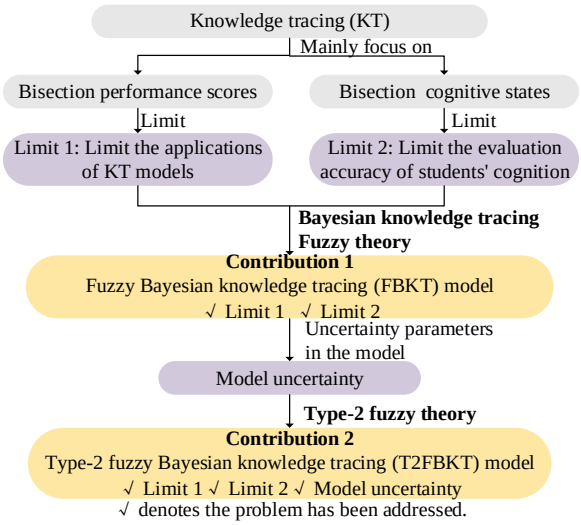

Fig. 1. Relationship between motivations and contributions in this study, where the purple and yellow boxes represent the limitations and contributions, respectively.

states and the relations between fuzzy cognitive states and fuzzy performance scores.

- Extensive experiments demonstrated that FBKT and T2FBKT achieve satisfactory performance in continuous score scenarios; moreover, they exhibit good explainability and identify subtler changes in student cognitive states. In addition, T2FBKT prevents model uncertainty to a certain degree compared with FBKT owing to the type-2 fuzzification.

This paper is organized as follows. Related work on KT models is reviewed in the next section. In Section III, background material is presented, including the formulation of the KT problem, the details of the traditional BKT model, and fuzzy theory. In Section IV, the two fuzzy KT models are proposed, namely FBKT and T2FBKT. In Section V the applications are presented. In Section VI the experiments are discussed. Finally, Section VII concludes the paper.

II. Related Work

Herein, we introduce KT models, including Bayesian [6]–[9] and deep-learning [10]–[13] KT models.

A. Bayesian KT Models

A representative Bayesian KT model is the BKT model proposed by Corbett et al. [6], which is detailed in Section III-B. This model has been improved in two aspects: a) by combining educational data features [17], [24]–[26], and b) by using mixed methods [21], [27]–[30].

Specifically, educational data features, such as knowledge components, exercises, time interval for exercises, and teaching intervention features, may be considered. The multi-grained-BKT and historical-BKT models with the knowledge hierarchy and historical information features, respectively, have been proposed for the Coursera MOOC platform [17]. Hawkins et al. [24] proposed the BKT-ST model considering the similarity of exercises between consecutive time steps. Temporal differences (related to exercise-completion) were integrated into the BKT model, resulting in the TD-BKT model [25]. Chen et al. [26] proposed an intervention-BKT model, assuming that various types of teaching interventions affect student cognition.

Regarding the extended models based on mixed methods, Zhang et al. [21] proposed the TLS-BKT model inspired by three-way decisions, improving the original unmastered/mastered states to unmastered/learning/mastering states so that the flexibility and universality of the BKT model could be enhanced. A dynamic Bayesian network (DBN) is applied to construct a knowledge hierarchy [27] in the KT model. In addition, [28], [29], and [30] combined the BKT model with IRT, HMM-IRT, and LFM, respectively.

Bayesian KT models model KT problems based on the HMM, the observation and hidden variables of which refer to performance and cognitive states, respectively. Student cognition can be easily tracked using the HMM. Hence, Bayesian KT models achieve strong interpretability.

B. Deep-learning KT Models

A representative deep-learning KT model is the deep knowledge tracing (DKT) model proposed by Piech et al. [10]. It is based on a recurrent neural network, which is a type of deep learning method. Deep-learning KT models can be classified into two categories: a) DKT variants [13], [31]–[33], and b) KT models based on other deep-learning methods [34], [35].

Specifically, regarding the former, Wang et al. [13] proposed the DTKS model, which integrates the relationships between exercises into the DKT model. Lalwani et al. [31] proposed the DKT–t model, which incorporates time interval as a feature of the DKT model. The DKT–t model can depict the forgetting curve of students. [32] and [33] proposed the PDKT–C and E2E–DKT models, which integrate the information related to knowledge and exercises, respectively.

Regarding the latter, Chen et al. [34] proposed the knowledge proficiency tracing (KPT), which associates a prior knowledge vector with each student and exercise. Memory-augmented neural networks (MANNs) were introduced into KT, resulting in the DKVMN model [35]. It has been demonstrated that DKVMN has a larger external storage space than the DKT model, and therefore it requires fewer parameters.

These KT models are based on deep learning methods, which have a strong presence of data. Therefore, they can make full use of these data to achieve satisfactory prediction performance.

III. Background

Herein, the KT problem is first formulated. Subsequently, the traditional BKT model and fuzzy theory are introduced. For clarity, notations are listed in order of appearance in Table I.

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TABLE I
TABLE OF NOTATIONS

Notations	Descriptions
KC_q	qth knowledge component;
E_i^t	Exercise answered by student i at time step t;
y_i^t	Observed performance score for exercise E_i^t answered by student i at time step t;
K_i^t	Knowledge component set related to exercise E_i^t answered by student i at time step t;
$k_{q,i}^t$	Cognitive state of KC_q of student i at time step t;
$\hat{y}_i^t$	Predicted performance score for exercise E_i^t answered by student i at time step t;
$P(L_t)$	Conditional probability that the student cognitive state is known at time step t;
$P(S), P(G)$	Slipping and guessing probabilities, respectively;
$A, \tilde{A}$	Classical and fuzzy sets, respectively;
N, M	Number of fuzzy cognitive states and fuzzy performance scores, respectively;
$\tilde{C}_i^t, \tilde{S}_i^t$	Fuzzy cognitive states and fuzzy performance scores of student i at time step t, respectively;
$\tilde{C}_n, \tilde{S}_m$	nth fuzzy cognitive states and mth fuzzy performance scores, respectively;
$\tilde{R}_1, \tilde{R}_2, \tilde{R}_3$	S2FP, FP2OP, and S2OP relations, respectively;
$\tilde{a}_{hn}$	Probability from the fuzzy cognitive state $\tilde{C}_h$ at the current time step to $\tilde{C}_n$ at the next time step;
$\tilde{b}_{nm}$	Probability from the fuzzy cognitive state $\tilde{C}_n$ to the fuzzy performance score $\tilde{S}_m$;
$\tilde{\eta}_m^{(1)}(y_t)$	Membership function expressing the degree to which the member of the observed performance score y_t belongs to the fuzzy performance score $\tilde{S}_m$;
$\tilde{\eta}_n^{(2)}(x_t)$	Membership function indicating the degree to which the member of the hidden cognitive state x_t belongs to the fuzzy cognitive state $\tilde{C}_n$;
$\tilde{\eta}_n^{(3)}(y_t)$	Membership function indicating the degree to which the member of the observed performance score y_t belongs to the fuzzy cognitive state $\tilde{C}_n$;
$\alpha_t^n, \beta_t^n, \gamma_t^n$	Forward, backward, and forward–backward variables for cognitive state C_n at time step t, respectively;
$\tilde{\alpha}_t^n, \tilde{\beta}_t^n, \tilde{\gamma}_t^n$	Fuzzy forward, backward, and forward–backward variables for the fuzzy cognitive state $\tilde{C}_n$ at time step t, respectively.

A. Formulation of the Knowledge Tracing Problem

We assume that a student answers an exercise individually at each time step. Moreover, I students work on exercises, and each exercise is related to one or more knowledge components. All the knowledge components are in the set $\{KC_q\}, q \in \{1, 2, \cdots, Q\}$. Specifically, student $i(i \in \{1, 2, \cdots, I\})$ answers exercise $E_i^t(t \in \{1, 2, \cdots, T\})$ at time step t. Exercise E_i^t is related to the knowledge component set $K_i^t(K_i^t \subseteq \{KC_q\}, q \in \{1, 2, \cdots, Q\})$. Before defining the KT model, we introduce two key concepts, known as the performance and cognitive state.

Definition 1. *Performance. The score y_i^t achieved by student i answering exercise E_i^t at time step t.*

Definition 2. *Cognitive state. The mastery degree $k_{q,i}^t$ of KC_q for student i at time step t.*

Knowledge tracing is defined as follows.

Definition 3. *Knowledge tracing. For observed performance scores $(y_i^1, y_i^2, \cdots, y_i^T)$ of student i from time step 1 to T, the objectives of KT are as follows. a) If K_i^t is provided, then the cognitive state $k_{q,i}^t$ can be tracked. The objective function is shown in Eq. (1). b) The performance score $\hat{y}_i^{T+1}$ at the future time step $(T+1)$ can be predicted. The objective function is shown in Eq. (2).*

$$\arg\max_k \left(\sum_{i=1}^{I} \sum_{q=1}^{Q} f\left(k_{q,i}^t; y_i^t, K_i^t\right) \right), \tag{1}$$

where $k = \{k_{q,i}^t\}, t \in \{1, 2, \cdots, T\}, q \in \{1, 2, \cdots, Q\}, i \in \{1, 2, \cdots, I\}$. $f\left(k_{q,i}^t; y_i^t, K_i^t\right)$ denotes the probability that student i achieves performance score y_i^t on exercise E_i^t as his cognitive state is $k_{q,i}^t$.

$$\arg\min_{\hat{y}^{T+1}} \left(\sum_{i=1}^{I} g\left(\hat{y}^{T+1}, y^{T+1}\right) \right), \tag{2}$$

where $\hat{y}^{T+1} = \{\hat{y}_i^{T+1}\}, i \in \{1, 2, \cdots, I\}$. $g\left(\hat{y}^{T+1}, y^{T+1}\right)$ denotes the error between the true performance score y^{T+1} and the predicted performance score $\hat{y}^{T+1}$.

B. Bayesian Knowledge Tracing

Bayesian KT models have attracted considerable attention owing to their strong interpretability. Herein, the details of the most representative BKT model [6] are presented as follows. It models student performance based on the HMM and estimates the probability of student cognitive states. The performance and cognitive states are described by bisection representations. The HMM describes the probability distribution of the performance scores $Y = (y_1, y_2, \cdots, y_T)$ over a discrete time series $(1, 2, \cdots, T)$, which depends on the state series $X = (x_1, x_2, \cdots, x_T)$. Y and X are known as the *observation variable* and *hidden variable*, respectively. Hence, the observation variable y_t at time step t depends only on the hidden variable x_t, and x_t only depends on x_{t-1}. In Fig. 2, (a) shows the framework of the BKT model from time step 1 to T, where the blue circles represent the hidden variables (i.e., cognitive states), whereas the red circles represent the observation variables (i.e., performance scores); (b) shows the relationship between the bisection cognitive states and bisection performance scores. According to the performance score $y_{q,t}$ related to KC_q at time step t, the conditional probability $P(L_t \mid y_{q,t})$ satisfies Eq. (3) if $y_{q,t} =$ right and satisfies Eq. (4) if $y_{q,t} =$ wrong.

$$P(L_t \mid y_{q,t}) = P(L_t)[1 - P(S)] + [1 - P(L_t)]P(G), \tag{3}$$

$$P(L_t \mid y_{q,t}) = P(L_t)P(S) + [1 - P(L_t)][1 - P(G)], \tag{4}$$

where $P(L_t)$ is the probability that the cognitive state is known at time step t. $P(S)$ and $P(G)$ are the slipping and guessing probabilities, respectively.

C. Fuzzy Theory

Herein, we discuss type-1 [36] and type-2 [37] fuzzy sets, fuzzy relations [38], and compositions of fuzzy relations [38].

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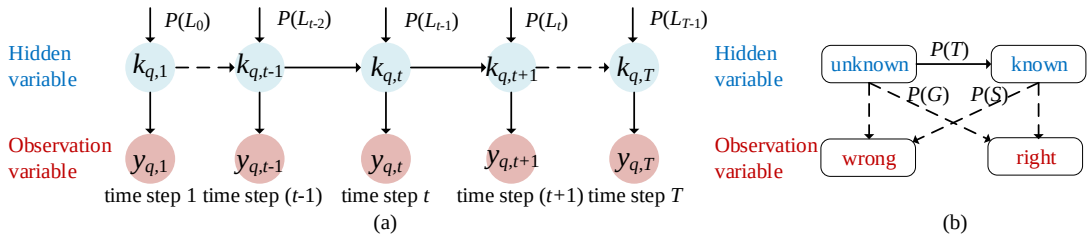

Fig. 2. Bayesian knowledge tracing model. (a) shows the framework of the BKT model from time step 1 to T, where the blue circles represent the hidden variables (i.e., cognitive states), whereas the red circles represent the observation variables (i.e., performance score). q denotes the qth knowledge component; $k_{q,t}$ and $y_{q,t}$ denote the hidden and observation variables related to the qth knowledge component at time step t, respectively; $P(L_0)$ and $P(L_t)$ denote the probability that the students know the qth knowledge component at time steps 0 and t, respectively. (b) shows the relationship between the bisection hidden and observation variables. $P(T)$, $P(G)$, and $P(S)$ denote the transition, guessing, and slipping probabilities, respectively.

*1) **Fuzzy Sets**:* Fuzzy sets are an extension of classical sets. In classical sets, only two relations exist between a member and a set: a) A member x belongs to a set A, and b) x does not belong to A. Therefore, the membership degree satisfies $\mu_A(x) \in \{0, 1\}$.

In type-1 fuzzy sets, member x belongs to fuzzy set $\tilde{A}$ with a certain probability. The membership degree satisfies $\mu_{\tilde{A}}(x) \in [0, 1]$.

A type-2 fuzzy set $\tilde{X}$ is defined on a domain Ω_X, and the membership $\mu_{\tilde{X}}(x)$ for a certain $x \in \Omega_X$ is a type-1 fuzzy set $U(x) = \mu_{\tilde{X}}(x)$ defined on $\Omega_{U(x)} \subseteq [0, 1]$ with membership function (MF) $\mu_{U(x)}(x, u_x)$, where $u_x \in \Omega_{U(x)} \subseteq [0, 1]$. $\mu_{\tilde{X}}(x)$ and $\mu_{U(x)}(x, u_x)$ are the primary and secondary MFs, respectively. The union of all primary MFs is recognized as the footprint of uncertainty (FOU). The FOU describes the uncertainty of the data, and the secondary MF describes the uncertainty of the primary MF. Type-2 fuzzy sets better represent the uncertainty that is due to the secondary MFs and FOU [23], [39].

*2) **Fuzzy Relations**:* As is the case with fuzzy sets, fuzzy relations are the extensions of classical relations. In classical relations, for two domains X and Y, the existence of a relation R between X and Y implies that either: a) $x \in X$ is related to $y \in Y$, or b) $x \in X$ is not related to $y \in Y$. A fuzzy relation $\tilde{R}$ between domains X and Y is denoted as $X \overset{\tilde{R}}{\sim\!\!\rightarrow} Y$, implying that $x \in X$ is related to $y \in Y$ with degree $\tilde{R}(x, y)$.

*3) **Composition of Fuzzy Relations**:* $X \overset{\tilde{T}}{\sim\!\!\rightarrow} Z$ is the composition of fuzzy relations $X \overset{\tilde{R}}{\sim\!\!\rightarrow} Y$ and $Y \overset{\tilde{S}}{\sim\!\!\rightarrow} Z$, written as $\tilde{T} = \tilde{R} \circ \tilde{S}$. The membership function of $\tilde{T}$ is expressed as as follows.

$$\tilde{T}(x, z) = \coprod_{y \in Y} (\tilde{R}(x, y) \prod \tilde{S}(y, z)), \forall (x, z), \tag{5}$$

where $\coprod$ and $\prod$ are the fuzzy union and intersection operations, respectively.

IV. Proposed Fuzzy Knowledge Tracing Models

Herein, the motivation for the proposed fuzzy KT models is first presented. Then, the framework of the fuzzy KT models is developed, and subsequently, FBKT and T2FBKT are described in detail. Finally, the time complexity of the traditional BKT model and the proposed models is analyzed.

A. Motivation

Existing KT models primarily focus on bisection representations [21], which limit the application of the KT model and the accuracy of the evaluation of student cognition, as shown in Fig. 3 (a). We distinguish the following four cases regarding the relation between bisection cognitive states and performance.

A student achieves the observed performance score of 0 in the following two cases: 1) He/she performs normally when his/her cognitive state is "unknown". 2) He/she makes a slip (make a mistake accidentally); however, his/her cognitive state is "known". By contrast, the student achieves the observed performance score of 1 in the following two cases: 1) He/she performs normally when his cognitive state is known. 2) He makes a guess; however, his/her cognitive state is "unknown". This type of model with limited representations can handle only these situations.

Fuzzy KT models have been proposed to address these limitations. As shown in Fig. 3 (b), the observed performance score (concept 1) and cognitive states (concept 2) are both fuzzified into M fuzzy performance scores and N fuzzy cognitive states. The relation between the fuzzy cognitive states and observed performance scores (relation 1) is composed of the relation between fuzzy cognitive states and fuzzy performance scores (relation 2) and the relation between fuzzy performance scores and observed performance scores (relation 3).

Therefore, fuzzy KT models are primarily concerned with the two fuzzy concepts and three fuzzy relations above.

B. Framework of Fuzzy Knowledge Tracing Models

As shown in Fig. 4, fuzzy KT models are concerned with two fuzzy concepts (namely, fuzzy cognitive states and fuzzy performance scores) and three fuzzy relations (namely, S2FP, FP2OP, and S2OP relations). To simplify the problem, it is assumed that a student answers an exercise at each time step, and an exercise is related to a knowledge component.

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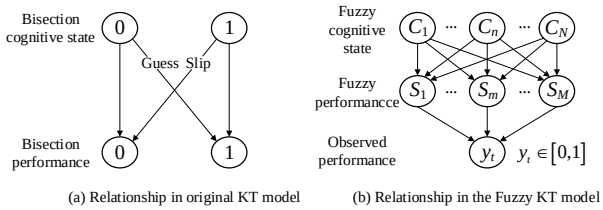

(a) Relationship in original KT model (b) Relationship in the Fuzzy KT model

Fig. 3. Relationships between cognitive states and performance scores in bisection and fuzzy representations.

*1) **Fuzzification of the Two Concepts**:* In this framework, the cognitive states and observed performance scores are fuzzified into N and M fuzzy sets, respectively. As shown in Fig. 4, at time step t, a student answers a question and achieves an observed performance score y_t, which may belongs to the fuzzy performance score $\tilde{S}_m \left(\tilde{S}_m \in \left\{\tilde{S}_1, \cdots, \tilde{S}_M\right\}\right)$. The fuzzy and observed performance scores refer to coarse and grained performance, respectively. At this time step, the student is in a fuzzy cognitive state $\tilde{C}_n \left(\tilde{C}_n \in \left\{\tilde{C}_1, \cdots, \tilde{C}_N\right\}\right)$ with a certain probability.

*2) **Fuzzification of the Three Relations**:* In this framework, the three relations refer to a) the relation from fuzzy cognitive states to fuzzy performance scores, defined as the S2FP relation (Definition 4), and b) the relation from fuzzy to observed performance scores, defined as the FP2OP relation (Definition 5); and c) the relation from the fuzzy cognitive states to observed performance scores, defined as the S2OP relation (Definition 6), which is composed of the S2FP and FP2OP relations.

Definition 4. *S2FP relation $\tilde{R}_1$. The fuzzy relation from the fuzzy cognitive state $\tilde{C}$ to the fuzzy performance score $\tilde{S}$ is denoted as $\tilde{C} \overset{\tilde{R}_1}{\sim\!\!\rightarrow} \tilde{S}$.*

Definition 5. *FP2OP relation $\tilde{R}_2$. The fuzzy relation from the fuzzy performance score $\tilde{S}$ to the observed performance score Y is denoted as $\tilde{S} \overset{\tilde{R}_2}{\sim\!\!\rightarrow} Y$.*

Definition 6. *S2OP relation $\tilde{R}_3$. The fuzzy relation from the cognitive state $\tilde{C}$ to the observed performance score Y is denoted as $\tilde{C} \overset{\tilde{R}_3}{\sim\!\!\rightarrow} Y, \tilde{R}_3 = \tilde{R}_1 \circ \tilde{R}_2$, where $\circ$ denotes composition of fuzzy relations.*

The S2FP relation $\tilde{R}_1$ indicates that a student in a good fuzzy cognitive state may not achieve good fuzzy performance owing to emotional factors (e.g., slipping, guessing, and gaming a system [40], [41]) and vice versa.

The FP2OP relation $\tilde{R}_2$ refers to defuzzification from coarse to fine-grained performance. We argue that an observed performance score belongs to more than one fuzzy performance score. For example, an observed performance score of 0.65 may belong to both $\tilde{S}_2$ and $\tilde{S}_3$ fuzzy performance scores.

The S2OP relation $\tilde{R}_3$ indicates that the observed performance is due to the fuzzy cognitive states of the students. We argue that the observed performance may be due to more than one fuzzy cognitive state with a certain probability. For example, an observed performance score of 0.65 may belong to the fuzzy cognitive state $\tilde{C}_3$ with a probability of 0.7, as well as to $\tilde{C}_4$ with a probability of 0.3.

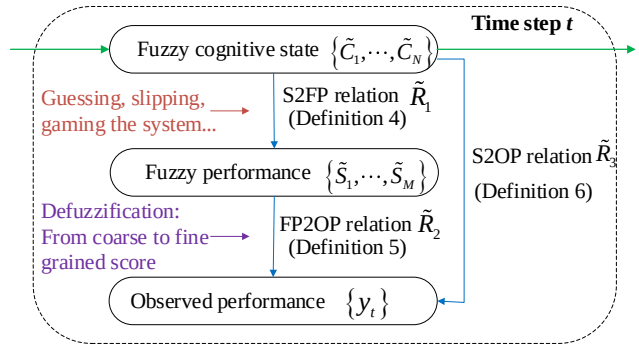

Fig. 4. Framework of the fuzzy KT models. The main task is to manage the two concepts and three relations. The two concepts refer to the fuzzy cognitive states and fuzzy performance scores. The three relations refer to the S2FP relation $\tilde{R}_1$, FP2OP relation $\tilde{R}_2$, and S2OP relation $\tilde{R}_3$.

By Eq. (5) in Section III-C, the membership function of the S2OP relation $\tilde{R}_3$ satisfies the following equation involving $\tilde{R}_1$ and $\tilde{R}_2$.

$$\begin{aligned} (\tilde{R}_3)(\tilde{C}_n, y_t) &= (\tilde{R}_1 \circ \tilde{R}_2)(\tilde{C}_n, y_t) \\ &= \coprod_{\tilde{S}_m \in \tilde{S}} (\tilde{R}_1(\tilde{C}_n, \tilde{S}_m) \prod \tilde{R}_2(\tilde{S}_m, y_t)), \\ \tilde{C}_n &\in \left\{\tilde{C}_1, \cdots, \tilde{C}_n, \cdots, \tilde{C}_N\right\}, y_t \in [0,1]. \end{aligned} \tag{6}$$

Hence, the framework is based on the fuzzification of the two concepts and the three relations.

C. Modeling

As described previously, the FBKT and T2FBKT models fuzzify two concepts and three relations. The difference between the two fuzzy models is that FBKT uses the type-1 fuzzy method in the fuzzification of both the concepts and relations, whereas T2FBKT uses the type-2 fuzzy method in the fuzzification of the relations (Fig. 5). As is the case with the traditional BKT model, FBKT and T2FBKT are modeled separately for each knowledge component.

*1) **Fuzzification of Two Concepts**:* The two concepts (i.e., performance scores and cognitive states) fuzzification of FBKT and T2FBKT using the type-1 fuzzy methods as follows.

We assume that the observed performance scores $(y_1, y_2, \cdots, y_T)$ are related to a knowledge component, where T is the total number of time steps. The observed performance scores are continuous, for example, from subjective examinations such as paper writing examinations. They are fuzzified into the M fuzzy performance scores $\tilde{S}_m, \left(\tilde{S}_m \in \left\{\tilde{S}_1, \tilde{S}_2, \cdots, \tilde{S}_M\right\}\right)$. We assume that the membership function $\tilde{\eta}_m^{(1)}(y_t)$ follows the Gaussian distribution, as shown in Eq. (7); this represents the probability that

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the observed performance score y_t belongs to the fuzzy performance score $\tilde{S}_m$.

$$\begin{aligned}\tilde{\eta}_m^{(1)}(y_t) &= \mathbb{N}\left(y_t;\mu_m^{(1)},\sigma_m^{(1)}\right)\\ &= \frac{1}{\sqrt{2\pi}\sigma_m^{(1)}}\exp\left(-\frac{\left(x-\mu_m^{(1)}\right)^2}{2\left(\sigma_m^{(1)}\right)^2}\right),\end{aligned} \tag{7}$$

where $\mu_m^{(1)}$ and $\sigma_m^{(1)}$ are the parameters of the fuzzy performance scores. It should be noted that an observed performance score may belong to multiple fuzzy performance scores with various probabilities.

The student cognitive states are known as hidden variables in the BKT model, denoted as $(x_1, x_2, \cdots, x_T)$. They are fuzzified into N fuzzy cognitive states. We assume that the membership function $\tilde{\eta}_n^{(2)}(x_t)$ also follows a Gaussian distribution, as shown in Eq. (8), which is similar to Eq. (7). $\tilde{\eta}_n^{(2)}(x_t)$ denotes the probability that the cognitive state x_t belongs to the fuzzy cognitive state $\tilde{C}_n$, $\left(\tilde{C}_n \in \left\{\tilde{C}_1, \tilde{C}_2, \cdots, \tilde{C}_N\right\}\right)$.

$$\tilde{\eta}_n^{(2)}(x_t) = \mathbb{N}\left(x_t;\mu_n^{(2)},\sigma_n^{(2)}\right), \tag{8}$$

where $\mu_n^{(2)}$ and $\sigma_n^{(2)}$ are parameters of the fuzzy cognitive states. It should be noted that a cognitive state may belong to multiple fuzzy cognitive states with various probabilities.

*2) **Fuzzification of the Three Relations**:* The three relations (i.e., S2FP relation $\tilde{R}_1$, FP2OP relation $\tilde{R}_2$, and S2OP relation $\tilde{R}_3$) are fuzzified by FBKT and T2FBKT using type-1 and type-2 fuzzy methods, respectively.

Specifically, $\tilde{R}_2$ (Definition 5) represents the transition from the fuzzy performance $\tilde{S}_m$ to observed performance score y_t, as shown in Eq. (7). $\tilde{R}_1$ (Definition 4) represents the transition from the fuzzy cognitive state $\tilde{C}_n$ to fuzzy performance score $\tilde{S}_m$. The probability of $\tilde{R}_1$ is denoted as $\tilde{b}_{nm}$, using the type-1 and type-2 fuzzy methods in FBKT and T2FBKT, respectively. As mentioned previously, $\tilde{R}_3$ is composed of $\tilde{R}_1$ and $\tilde{R}_2$. As is the case with $\tilde{R}_1$, $\tilde{R}_3$ is fuzzified using the type-1 and type-2 fuzzy methods in FBKT and T2FBKT, respectively. The assumptions regarding $\tilde{R}_1$ and $\tilde{R}_2$ are as follows.

Assumption 1. *The probability of the S2FP relation $\tilde{R}_1$ is the degree of membership of a fuzzy performance score belonging to a fuzzy cognitive state.*

Assumption 2. *The probability of the FP2OP relation $\tilde{R}_2$ is the degree of membership of an observed performance score belonging to a fuzzy performance score.*

The S2OP relation $\tilde{R}_3$ (Definition 6) represents the transition from the fuzzy cognitive state $\tilde{C}_n$ to the observed performance y_t. According to the composition function in Eq. (6), the membership function $\tilde{\eta}_n^{(3)}(y_t)$ is as follows.

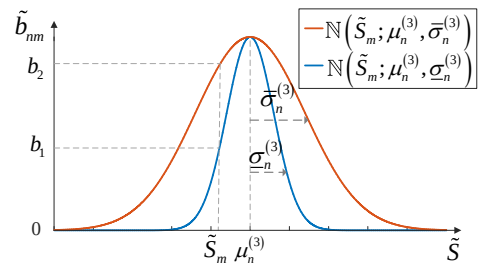

Fig. 5. Membership function $\tilde{b}_{nm}$ of the type-2 fuzzification of the S2FP relations in T2FBKT. The probability that the fuzzy performance score $\tilde{S}_m$ belongs to fuzzy cognitive state $\tilde{C}_n$ is in the range $[b_1, b_2]$.

$$\begin{aligned}\tilde{\eta}_n^{(3)}(y_t) &= \coprod_{\tilde{S}_m\in\tilde{S}}\left(\tilde{R}_1(\tilde{C}_n,\tilde{S}_m)\prod\tilde{R}_2(\tilde{S}_m,y_t)\right)\\ &= \coprod_{\tilde{S}_m\in\tilde{S}}\left(\tilde{b}_{nm}\prod\tilde{\eta}_m^{(1)}(y_t)\right),\\ &\tilde{C}_n\in\left\{\tilde{C}_1,\cdots,\tilde{C}_n,\cdots,\tilde{C}_N\right\}, y_t\in[0,1].\end{aligned} \tag{9}$$

$\tilde{\eta}_m^{(1)}(y_t)$ denotes the probability of the FP2OP relation $\tilde{R}_2$, which is given in Eq. (7). $\tilde{b}_{nm}$ denotes the probability of the S2FP relation $\tilde{R}_1$. We assume that $\tilde{b}_{nm}$ follows a Gaussian distribution according to the type-1 fuzzy method in the FBKT model (Eq. (10)), and the type-2 fuzzy method in the T2FBKT model (Eq. (11)).

$$\tilde{b}_{nm} = \tilde{\mathbb{N}}\left(\tilde{S}_m;\mu_n^{(3)},\sigma_n^{(3)}\right), \tag{10}$$

where $\mu_n^{(3)}$ and $\sigma_n^{(3)}$ are parameters of the S2FP relation. $\sigma_n^{(3)}$ is a fixed value according to the type-1 fuzzy method.

$$\begin{aligned}&\tilde{b}_{nm} = \tilde{\mathbb{N}}\left(\tilde{S}_m;\mu_n^{(3)},\sigma_n^{(3)}\right), \sigma_n^{(3)}\in\left[\underline{\sigma}_n^{(3)},\overline{\sigma}_n^{(3)}\right],\\ &\overline{\sigma}_n^{(3)} = k_1*\sigma_n^{(3)}, \underline{\sigma}_n^{(3)} = k_2*\sigma_n^{(3)}.\end{aligned} \tag{11}$$

Unlike in Eq. (10), $\sigma_n^{(3)}$ is in the range $\left[\underline{\sigma}_n^{(3)},\overline{\sigma}_n^{(3)}\right]$ in Eq. (11) according to the type-2 fuzzy method. For example, as shown in Fig. 5, $\tilde{b}_{nm}$ is in the range $[b_1, b_2]$. k_1 and k_2 are the upper and lower factors of $\sigma_n^{(3)}$, respectively.

In summary, in the modeling of FBKT and T2FBKT, the observed performance scores of a student, which are related to one knowledge component, are collected. First, the two concepts are fuzzified into M fuzzy performance scores and N fuzzy cognitive states. Subsequently, the three relations are fuzzified using type-1 and type-2 fuzzy methods in FBKT and T2FBKT, respectively. Finally, these fuzzy components are modeled as HMMs in the form of the traditional BKT model.

D. Training

Herein, we introduce the training algorithm for FBKT and T2FBKT. The forward–backward algorithm [23], a well-known training algorithm for HMMs, is adapted to the proposed models.

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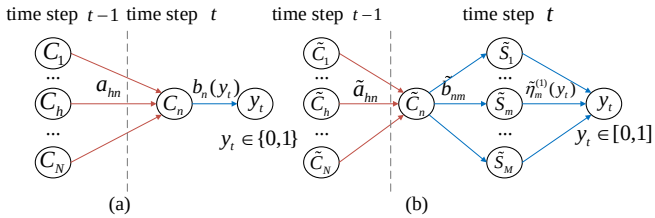

Fig. 6. Forward variable maps in the traditional forward–backward algorithm (a) and fuzzy forward–backward algorithm (b).

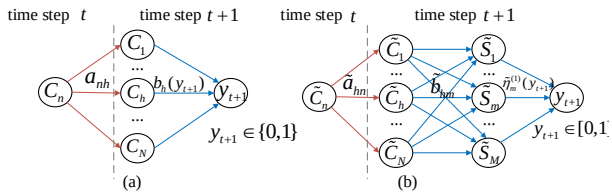

Fig. 7. Backward variable maps in the traditional forward–backward algorithm (a) and fuzzy forward–backward algorithm (b).

In the forward–backward algorithm [23], the forward and backward variables are defined in Eqs. (12) and (13), respectively (as shown in Figs. 6(a) and 7(a)). The forward–backward variable is defined in Eq. (14).

$$\alpha_t(n) = P(y_1, y_2, \cdots, y_t, i_t = C_n \mid \lambda), \tag{12}$$

$$\beta_t(n) = P(y_{t+1}, y_{t+2}, \cdots, y_T \mid i_t = C_n, \lambda), \tag{13}$$

$$\gamma_t(n) = P(i_t = C_n \mid y_1, y_2, \cdots, y_T, \lambda), \tag{14}$$

where $\lambda =< \Pi, A, B >$ is the HMM parameter. i_t and y_t represent the hidden and observation variables at time step t, respectively. $\alpha_t(n)$, $\beta_t(n)$, and $\gamma_t(n)$ are calculated as shown in Eqs. (15), (16), and (17), respectively.

$$\alpha_t(n) = \begin{cases} \pi_n b_n(y_t), & t = 1 \\ \left[\sum_{h=1}^{N} \alpha_t(h) a_{hn}\right] b_n(y_t), & t \geqslant 2 \end{cases} \tag{15}$$

$$\beta_t(n) = \begin{cases} \sum_{h=1}^{N} a_{nh} b_h(y_{t+1}) \beta_{t+1}(h), & t \leq T-1 \\ 1, & t = T \end{cases} \tag{16}$$

$$\gamma_t(n) = \alpha_t(n) * \beta_t(n), \tag{17}$$

where $a_{hn} \in A$ denotes the probability of the transition from state C_h to C_n, $b_n(y_t) \in B$ denotes the probability of the transition from state C_n to the observation y_t, and $\pi_n \in \Pi$ is the probability of the initial hidden state C_n.

In the proposed models, hidden and observation variables refer to fuzzy cognitive states and observed performance scores, respectively. As shown in Figs. 6(b) and 7(b), the fuzzy forward, fuzzy backward, and fuzzy forward–backward variables are adapted in Eqs. (18), (19), and (20), respectively.

$$\tilde{\alpha}_t(n) = \begin{cases} \tilde{\eta}_n^{(2)}(x_t) \prod \tilde{\eta}_n^{(3)}(y_t), t = 1 \\ \left\{\coprod_{h=1}^{N} [\tilde{\alpha}_{t-1}(h) \prod \tilde{a}_{hn}]\right\} \prod \tilde{\eta}_n^{(3)}(y_t), \quad t \geqslant 2 \end{cases} \tag{18}$$

$$\tilde{\beta}_t(n) = \begin{cases} \coprod_{h=1}^{N} \left[\tilde{a}_{nh} \prod \tilde{\eta}_h^{(3)}(y_{t+1}) \prod \tilde{\beta}_{t+1}(h)\right], & t \leq T-1 \\ 1, & t = T \end{cases} \tag{19}$$

$$\tilde{\gamma}_t(n) = \tilde{\alpha}_t(n) \prod \tilde{\beta}_t(n), \tag{20}$$

By Eq. (9), $\eta_n^{(3)}(y_t) = \coprod_{m=1}^{M} \left[\tilde{b}_{nm} \prod \tilde{\eta}_m^{(1)}(y_t)\right]$. $\tilde{a}_{hn}$ denotes the probability of the transition from the fuzzy cognitive state $\tilde{C}_h$ to $\tilde{C}_n$.

During training, $\tilde{a}_{hn}$, $\mu_n^{(3)}$, and $\left[\sigma_n^{(3)}\right]^2$ are updated as in Eqs. (21), (22), and (23), respectively.

$$\tilde{a}_{hn} = \frac{\coprod_{t=1}^{T-1} \coprod_{m=1}^{M} P(O, j_t = \tilde{S}_m, i_t = \tilde{C}_h, i_{t+1} = \tilde{C}_n \mid \hat{\lambda})}{\coprod_{t=1}^{T-1} \coprod_{m=1}^{M} P(O, j_t = \tilde{S}_m, i_t = \tilde{C}_h \mid \hat{\lambda})}. \tag{21}$$

In Eq. (21), the numerator refers to the expectation that the fuzzy cognitive states are $\tilde{C}_h$ and $\tilde{C}_n$ at two consecutive time steps. The denominator refers to the expectation that the fuzzy cognitive state is $\tilde{C}_h$ at any time step.

$$\mu_n^{(3)} = \frac{\coprod_{m=1}^{M} \left[\tilde{S}_m \prod p_n(m)\right]}{\coprod_{m=1}^{M} p_n(m)}. \tag{22}$$

In Eq. (22), the numerator refers to the total expectation that the fuzzy performance score is $\tilde{S}_m, m \in \{1, 2, ..., M\}$ with probability $p_n(m)$ when the fuzzy cognitive state is $\tilde{C}_n$. The denominator refers to the total expectation of probability $p_n(m)$, which satisfies Eq. (24).

$$\left[\sigma_n^{(3)}\right]^2 = \frac{\coprod_{m=1}^{M} \left[\left(\tilde{S}_m - \mu_n^{(3)}\right)^2 \prod p_n(m)\right]}{\coprod_{m=1}^{M} p_n(m)}, \tag{23}$$

where $\mu_n^{(3)}$ and $\sigma_n^{(3)}$ are the parameters in $\eta_h^{(3)}(y_t)$.

$$p_n(m) = \frac{\coprod_{t=1}^{T} \left[P(O, m_t = \tilde{S}_m, i_t = \tilde{C}_n \mid \hat{\lambda})\right]}{\coprod_{t=1}^{T} P(O, i_t = \tilde{C}_n \mid \hat{\lambda})}. \tag{24}$$

In Eq. (24), the numerator refers to the expectation that the fuzzy performance score is $\tilde{S}_m$, and the fuzzy cognitive state is $\tilde{C}_n$. The denominator refers to the expectation that the fuzzy cognitive state is $\tilde{C}_n$.

The log-likelihood function of the proposed models is as follows. The derivation is presented in Appendix A.

$$\hat{\lambda} = \arg\max_{\lambda} \coprod_{N} \coprod_{M} P(Y, I, J \mid \hat{\lambda}) \log \tilde{\eta}_{i_1}^{(2)}(x_1) \coprod \left[\coprod_{t=1}^{T} \log \tilde{b}_{i_t, j_t} \coprod_{t=1}^{T} \log \tilde{\eta}_{j_t}^{(1)}(y_t) \coprod_{t=1}^{T-1} \log \tilde{a}_{i_t i_{t+1}}\right], \tag{25}$$

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where λ is a model parameter. Y, I, and J denote the observed performance scores, fuzzy cognitive states, and fuzzy performance scores, respectively.

E. Time complexity

Let N and M denote the number of hidden and observed variables, respectively. Let S denote the number of students, and let T denote the maximum length of the observed performance scores.

1) Time complexity of the traditional BKT model: In the traditional BKT model, we have $N = 2$. According to the training algorithm of the BKT model, that is, the forward–backward algorithm, for each epoch, the parameters are updated after the forward and backward algorithms run for the observed performance scores of all students. As shown in Fig. 6, in each time step, there are N^2 state transitions. The time complexity of the forward and backward algorithms is TN^2. Therefore, the time complexity of the traditional BKT model is $O(STN^2)$.

2) Time complexity of the fuzzy KT models: The time complexity of the proposed fuzzy KT models depends on that of the fuzzy forward and backward algorithms. As shown in Fig. 7, in each time step, there are N^2M transitions from fuzzy cognitive states to observed performance scores. The time complexity of the proposed models is $O(STN^2M)$, where S is the number of students. T is the maximum number of time steps for the observed performance scores for one student. N and M are the number of fuzzy performance scores and fuzzy cognitive states, respectively.

V. Applications

The proposed FBKT and T2FBKT model student performances during the learning process. They evaluate student cognitive states as time progresses, and predict future performance accordingly.

In contrast to the traditional BKT models, the proposed models identify student cognitive states more accurately. This is because cognitive states are fuzzified into several fuzzy sets; hence, continuous cognitive states, which are more accurate than discrete states, can be obtained. Furthermore, the proposed models address subjective problems and achieve better prediction performance because the observed performance scores are fuzzified into several fuzzy performance scores. Therefore, the proposed models have wider applicability.

A. Analysis of Student Cognition

For the cognitive states of student i pertaining to KC_q, we collect student performance scores on the exercises related to KC_q, denoted by $(y_1, y_2, \cdots, y_T)$. Subsequently, these scores are modeled, and the forward–backward variable sequence $\left(\tilde{\mathbf{\Gamma}}_1, \tilde{\mathbf{\Gamma}}_2, \cdots, \tilde{\mathbf{\Gamma}}_T\right)$ is iteratively obtained, in which $\tilde{\mathbf{\Gamma}}_t = \{\tilde{\gamma}_t(1), \tilde{\gamma}_t(2), \cdots, \tilde{\gamma}_t(n), \cdots, \tilde{\gamma}_t(N)\}$. According to the definition in Subsection B, the forward–backward variable $\tilde{\gamma}_t(n)$ with a continuous value refers to the probability that the cognitive state belongs to $\tilde{C}_n$ for $(y_1, y_2, \cdots, y_T)$ and parameter λ is shown. The cognitive state of the student belongs to the fuzzy cognitive states $\tilde{C}_1, \cdots, \tilde{C}_N$ with probabilities $\{\tilde{\gamma}_t(1), \tilde{\gamma}_t(2), \cdots, \tilde{\gamma}_t(n), \cdots, \tilde{\gamma}_t(N)\}$ at time step t. Hence, the continuous cognitive state x_t at time step t can be obtained by defuzzifying $\tilde{\eta}_n^{(2)}(x_t)$ (Eq. (8)). Therefore, the cognitive states of the student can be evaluated.

B. Prediction of Student Performance

Based on the analysis above, future student performance can be predicted. For example, $\tilde{\mathbf{\Gamma}}_T = \{\tilde{\gamma}_T(1), \tilde{\gamma}_T(2), \cdots, \tilde{\gamma}_T(n), \cdots, \tilde{\gamma}_T(N)\}$ is obtained for the cognitive states of student i pertaining to KC_q. Hence, we can simulate $\tilde{\mathbf{\Gamma}}_{T+1}$ through the probability $\tilde{a}_{nh}$ of the transition from the fuzzy cognitive state $\tilde{C}_n$ to $\tilde{C}_h$. Therefore, the probability that the fuzzy cognitive state at time step $T+1$ is $\tilde{C}_h$ satisfies

$$\tilde{\gamma}_{T+1}(h) = \coprod_{n=1}^{N} \left[\tilde{\gamma}_T(n) \prod \tilde{a}_{nh}\right]. \tag{26}$$

where $\tilde{a}_{nh}$ denotes the transition probability from the fuzzy cognitive state $\tilde{C}_n$ to $\tilde{C}_h$. Subsequently, the probabilities $\Psi_{T+1} = \left\{\tilde{\psi}_{T+1}(1), \tilde{\psi}_{T+1}(2), ...\tilde{\psi}_{T+1}(m), ...\tilde{\psi}_{T+1}(M)\right\}$ of the fuzzy performance scores $\tilde{S}_1, \tilde{S}_2, ..., \tilde{S}_m, ..., \tilde{S}_M$ are calculated based on $\tilde{\mathbf{\Gamma}}_{T+1}$ according to Eq. (27).

$$\tilde{\psi}_{T+1}(m) = \coprod_{n=1}^{N} \left[\tilde{\gamma}_{T+1}(h) \prod \tilde{b}_{nm}\right] \tag{27}$$

where $\tilde{b}_{nm}$ denotes the transition probability from the fuzzy cognitive state $\tilde{C}_n$ to the fuzzy performance score $\tilde{S}_m$. Here, the continuous prediction performance $\hat{y}_{T+1}$ at time step $T+1$ is obtained by the defuzzification process based on $\tilde{\mathbf{\Psi}}_{T+1}$ according to Eq. (28).

$$\hat{y}_{T+1} = \coprod_{m=1}^{M} \left[\psi_{T+1}(m) \prod \tilde{\eta}_m^{(1)}\right] \tag{28}$$

where $\tilde{\eta}_m^{(1)}$ is the membership function following the Gaussian distribution as shown in Eq. (7).

In conclusion, the prediction performance can be obtained using the proposed models, based on the cognitive states of the students.

VI. Experiments

Herein, we describe extensive experiments that were conducted to demonstrate the effectiveness of FBKT and T2FBKT. We first present the experimental setting. Subsequently, the fuzzy KT models are compared with 13 baseline models. Finally, the results are discussed, particularly the explainability of the proposed models.

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A. Preliminaries

Herein, the experimental setting is presented, including the datasets, baselines, and metrics.

*1) **Datasets**:* The experiments were conducted on the following six well-known real-world datasets. The first three datasets are from the KDD Cup 2010 Educational Data Mining Challenge. The other datasets are from the adaptive mathematics tutor ASSISTments for different school years. The data format can be found on the homepage.

- Algebra I 2005–2006 [42], denoted as Algebra05 [a].
- Algebra I 2006–2007 [43], denoted as Algebra06 [a].
- Bridge to Algebra 2006–2007 [44]: Written as Bridge_algebra06 [a].
- ASSISTments 2009-2010 [45]: From the adaptive mathematics tutor ASSISTments (2009–2010 school year), denoted as ASSISTments09 [b].
- ASSISTments 2012-2013: From the adaptive mathematics tutor ASSISTments (2012–2013 school year), denoted as ASSISTments12 [c].
- ASSISTments 2017: From the 2017 ASSISTments Longitudinal Data Mining Competition, denoted as ASSISTments17 [d].

As in existing studies on KT problem [14], [21], in the experiments, we filtered all students who practiced less than 10 exercises. Although the proposed models can be applied to the datasets without this preprocessing, the modification has been adopted to maintain consistency with the data preprocessing in other studies [14], [21] and ensure the reliability of the results of all KT models [14], [21].

Subsequently, to evaluate the performance of the proposed models, data preprocessing was performed to obtain the continuous scores referred to in [46], [47], which are based on data attributes, such as, the score at the first action, whether the students asks for hints, and whether the student attempts for more than one time. The dataset preprocessing is detailed in Part I of Supplements.

After preprocessing, the composition of the continuous scores for the six datasets is provided in Fig. 8. Specifically, in each pie, the score distribution is shown. The information regarding each dataset is shown in Table II [e]

*2) **Baselines**:* The proposed models were compared with the following algorithms to demonstrate the effectiveness: BKT [6], DKT [10], DKVMN [35], DKT+ [48], DKT_DSC [49], DeepIRT [50], SAKT [51], KQN [52], AKT [53], KQN [52], AKT [53], CKT [54], DMN [55], IADMN [55], and IADMN [55].

[a]https://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp

[b]https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/

[c]https://sites.google.com/site/assistmentsdata/home/2012-13-school-data-with-affect/

[d]https://sites.google.com/view/assistmentsdatamining/

[e]"Ques" is an abbreviation of "questions." "KC" is an abbreviation of "knowledge components". "Proportion" indicates "proportion of continuous scores". Note that the continuous scores refer to the scores excluding 0 and 1.0. Note that continuous scores are scores other than 0 and 1.

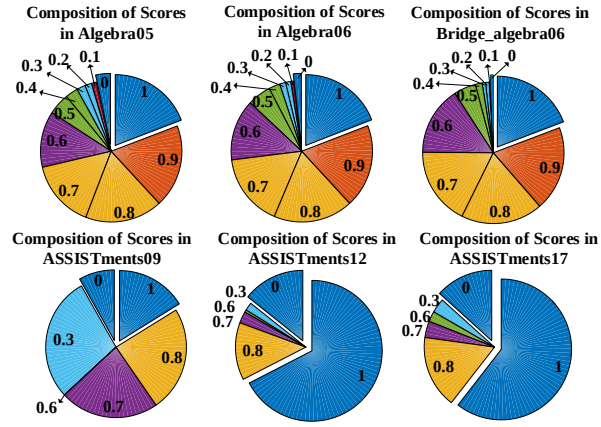

Fig. 8. Composition of continuous scores for the six datasets after preprocessing. In each pie, the score distribution is shown.

TABLE II
PREVIEW OF DATASETS [e]

Dataset	No_Users	No_Ques	No_KC	No_Logs	Proportion
Algebra05	546	164,401	108	549,849	77.4%
Algebra06	1,279	475,472	289	1,567,377	79.0%
Bridge_algebra06	1,112	116,908	317	1,601,985	80.1%
ASSISTments09	3,062	4,371	83	162,690	75.6%
ASSISTments12	16,587	47,366	142	1,683,393	18.3%
ASSISTments17	1,704	3,092	79	771,378	26.2%

TABLE III
PARAMETER SETTINGS

Parameter	Value
Validation method	5-fold
Epochs in each fold	100
Precision for iteration termination	0.001
Number of fuzzy cognitive states	5
Number of fuzzy performance scores	5

*3) **Metrics**:* The following two metrics were selected to quantify prediction performance [14] from a regression perspective. Smaller values indicate better performance.

- **RMSE**: Root mean squared error.
- **MAE**: Mean absolute error.

These refer to the L2 and L1 norms, respectively. Hence, RMSE is more sensitive to outliers, as it would be large if there were an outlier with a large difference between the predicted and true value.

In the following, we compare the proposed models with the baselines in terms of performance prediction using non-parametric statistical tests. Subsequently, we study the effect of model parameters, namely, the number of fuzzy cognitive states, the number of fuzzy performance scores, and the range of k in Eq. (11), which denotes the type-2 fuzzy amplitude of $\sigma_n^{(3)}$. Finally, we discuss the results, particularly regarding the explainability of the proposed models.

B. Performance Prediction Comparisons

The proposed fuzzy KT models (i.e., FBKT and T2FBKT) were compared with 13 KT models. Some parameters in

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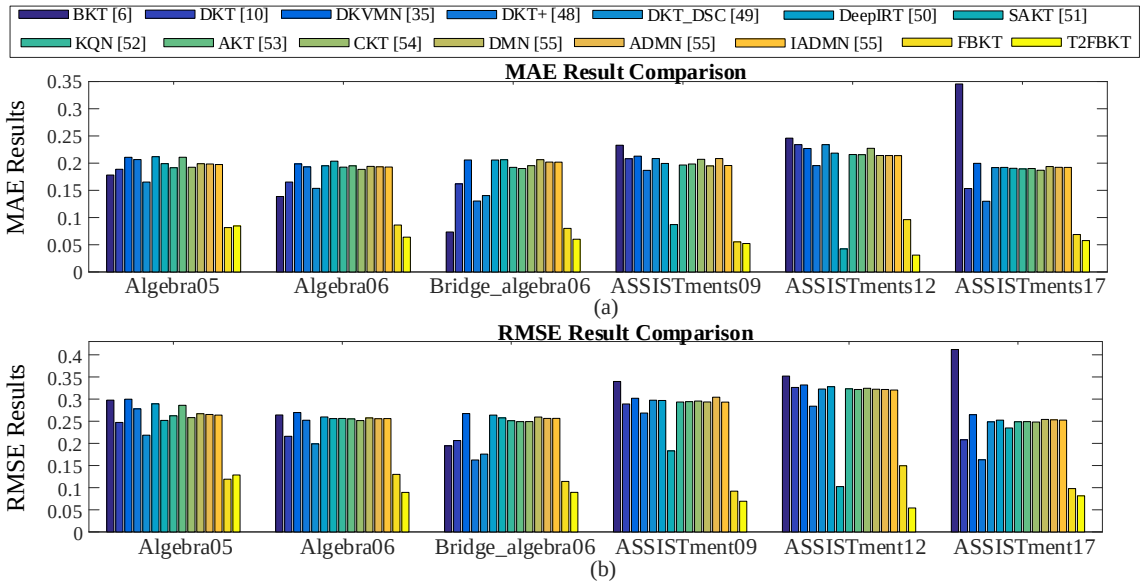

Fig. 9. Prediction results for the proposed models and baselines on six datasets. The fuzzy KT models achieve satisfactory performance in continuous score scenarios.

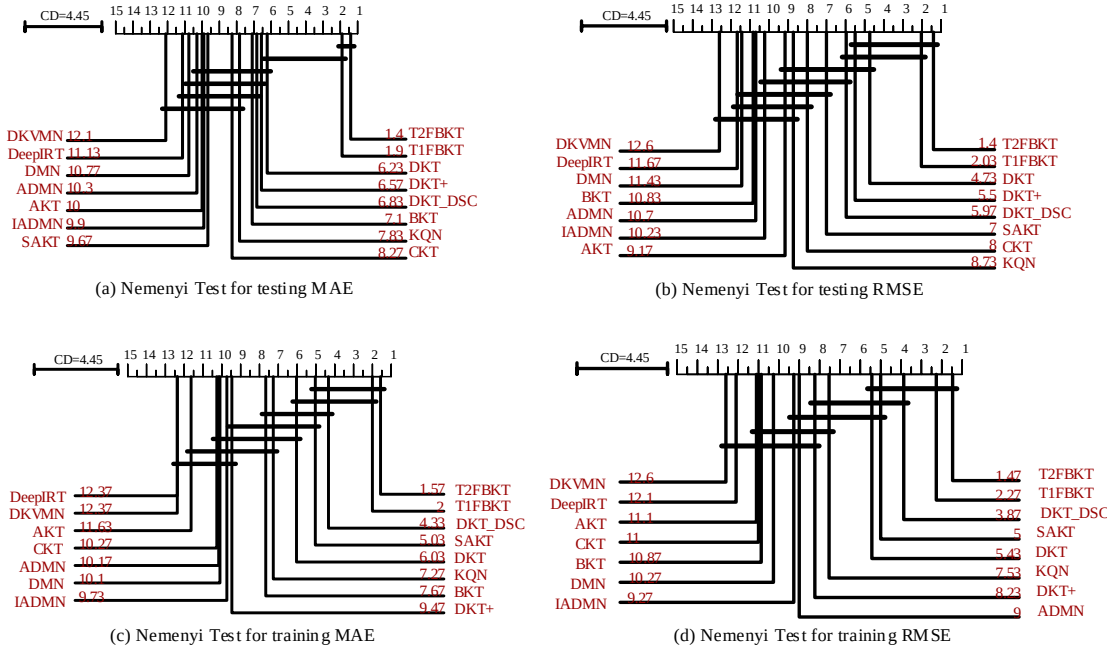

Fig. 10. Nemenyi test results for the four metrics, i.e., the RMSE results in the training and testing data, and the MAE results in the training and testing data. The average rank of each algorithm is marked along the axis (lower ranks to the right). The models on the same horizontal line have a similar prediction performance. In the continuous scenario, the proposed model is observed to be significantly better than most compared models.

the models are listed in Table III. To ensure fairness, the parameters *epoch in each fold* and *precision for iteration termination* in the baseline models are the same as those in the proposed models.

1) ***Comparisons****:* All the experiments were conducted using the *5-fold cross validation* method to obtain stable results. For comparison, the best performance was recorded for each model during the iterations. It should be noted that we treated partially correct responses as wrong because of the inapplicability of subjective problems [22].

As shown in and Fig. 9, the prediction performance (including the average RMSE and MAE results) of the FBKT and T2FBKT models is significantly better than that of the baseline models. This indicates that the fuzzy models can make full use of the continuous scores. Hence, the fuzzy cognitive states and score sets in the fuzzy models

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are conducive to the prediction performance. Moreover, T2FBKT outperforms FBKT in most cases, indicating that fuzzy KT models using the type-2 fuzzy method avoid model uncertainty to a certain degree, compared with those using the type-1 fuzzy method (the values of prediction performance are shown in Part III of Supplements).

*2) **Non-parametric statistical tests:*** To provide a comprehensive performance comparison between the proposed FBKT and T2FBKT models and other KT models, the Friedman and Nemenyi tests [56] were conducted in the experiments, which are widely used to statistically compare different algorithms over multiple datasets.

The Friedman test is used to determine whether the proposed models and baselines have the same performance. It is measured by calculating the average ranks, where a smaller rank indicates better performance. As shown in Fig. 10, the average ranks of the fuzzy BKT models in the four metrics (i.e., the training RMSE and MAE, and the testing RMSE and MAE) are the best in the continuous scenario. The average rank of the T2FBKT model is better than that of the FBKT model, which indicates that the type-2 theory plays a role in the performance of the fuzzy KT models.

In the Nemenyi tests, it is considered that a significant difference exists if the average ranks of two models differ by at least one critical difference (CD), which is calculated using a 5% significance level. The CD diagrams for the four metrics are plotted in Fig. 10, where the average rank of each algorithm is marked along the axis (lower ranks to the right). In Fig. 10, the models on the same horizontal line have similar prediction performance. In the continuous scenario, the proposed model is observed to be significantly better than most compared models.

C. Discussion

The results indicate that FBKT and T2FBKT outperform the baseline models. In particular, T2FBKT avoids model uncertainty to a certain degree compared with FBKT. The superiority of the proposed models may be attributed to the following: 1) The fuzzy KT models make full use of the continuous score information in the datasets through the score fuzzification. 2) The fuzzy KT models trace subtler changes in student cognition through the fuzzification of cognitive states. Here, we analyze student cognitive states in detail to study the explainability of the proposed models.

Explainability of continuously changing cognitive states for two consecutive time steps: Between the two consecutive time steps, the fuzzy KT models explain the performance based on the fine-grained changes in the cognitive states of students (Fig. 11 (d-f)). However, most existing KT models adopt bisection representations of the performance scores and cognitive states (Fig. 11 (a-c)), which generates the following two types of limitations.

Case 1: A student received significantly declining/rising scores [f] at two consecutive time steps: i) For this type of case, most traditional KT models usually regard the students' cognitive states as known→unknown or unknown→known. In addition, the traditional KT models may regard them as known→known or unknown→unknown because there are guessing and slipping probabilities in the KT models. Therefore, certain partially known states are ignored during the human learning process. ii) However, the students' cognitive states by the fuzzy KT models are regarded as continuously declining or rising (e.g., 0.42→0.24 (Example 1-1) or 0.31→0.70 (Example 1-2)), not known→unknown or unknown→known. The fuzzy KT models consider that a high score does not necessarily imply that the student has a complete mastery of the knowledge component, and vice versa.

Example 1-1: As shown in Fig. 11, the student received significantly declining scores (0.5→0.05) at time steps 8 to 9 (Case 1-1 in (a)). In the traditional KT models, the scores were described as 1→0 (Case 1-1 in (d)). Through the cognitive state evaluation (Case 1-1 in (f)), it is misunderstood that the student knows nothing about the knowledge component at time step 9. However, through the cognitive state evaluation of the fuzzy KT models, the student is determined to have decreased cognitive states (0.42→0.24) from time steps 8 to 9 (Case 1-1 in (c)) according to the original significantly declining scores (Case 1-1 in (a)).

Example 1-2: As shown in Fig. 11, the student received the significantly rising scores (0.3→0.6) at time steps 23 to 24 (Case 1-2 in (a)). In the traditional KT models, the scores were described as 0→1 (Case 1-2 in (d)). Through the cognitive state evaluation (Case 1-2 in (f)), it is misunderstood that the student has no cognition improvement after receiving the rising scores. However, through the cognitive state evaluation of the fuzzy KT models, it is determined that the student has increased cognitive states (0.31→0.70) from time steps 23 to 24 (Case 1-2 in (c)) according to the original significantly rising scores (Case 1-2 in (a)).

Case 2: A student received slightly declining/rising scores at two consecutive time steps: i) For this type of case, most traditional KT models usually regard the students' cognitive states as unknown→unknown (Example 2-1) or known→known (Example 2-2). Therefore, certain partially known states are ignored during the human learning process. ii) However, the students' cognitive states by the fuzzy KT models are regarded as a slight decrease/increase (e.g., 0.27→0.20 (Example 2-1) or 0.69→0.76 (Example 2-2)), and not as unknown→unknown or known→known. The fuzzy KT models consider that when a student receives slightly declining scores, their cognitive states are not constant, but may also have a slight decline, and vice versa.

Example 2-1: As shown in Fig. 11, the student received slightly declining scores (0.4→0.3) at time steps 11 to 12 (Case 2-1 in (a)). In the traditional KT models, the scores were described as 0→0 (Case 2-1 in (d)). Through the cognitive state evaluation (Case 2-1 in (f)), it is misunderstood that the student has no cognitive change when he provides consecutive incorrect answers. However, through

[f] All observation performance scores were normalized to values belonging to [0,1].

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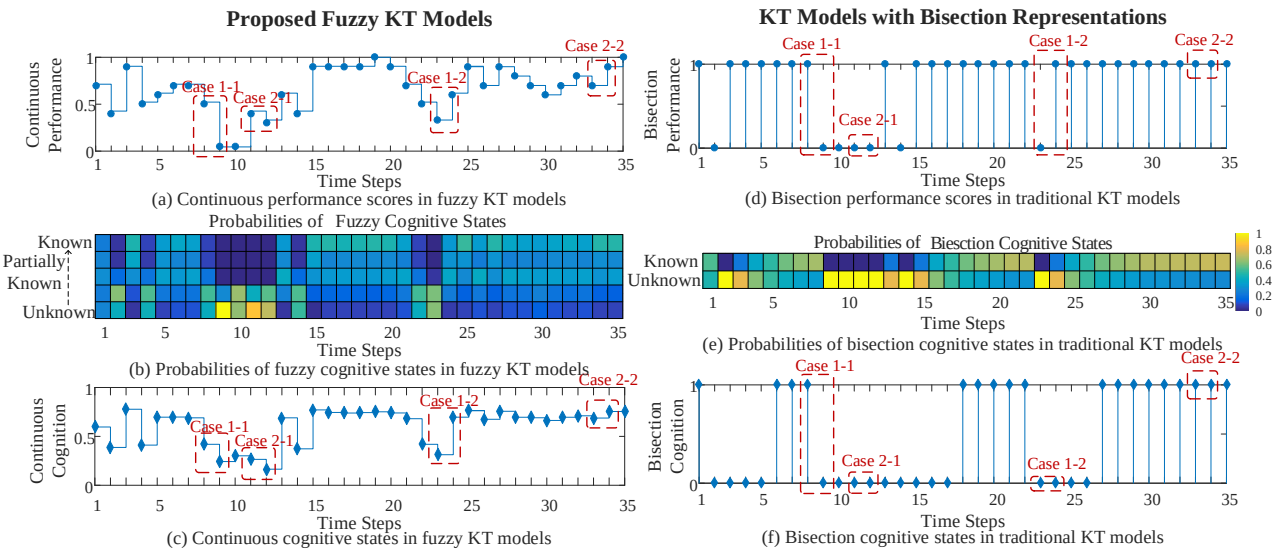

Fig. 11. Example to describe the explainab ility of the proposed fuzzy KT models compared with existing KT models using bisection representations. The fuzzy KT models can address two types of cases that cannot be handled by existing KT models. In the experiment, it is assumed that a cognitive state may transit from "known" to "unknown." In the figure, (a), (b), and (c) show the cognitive performance, probabilities of fuzzy cognitive states, and continuous cognitive states in the fuzzy KT models, respectively; (d), (e), and (f) show the bisection performance, probabilities of bisection cognitive states, and bisection cognitive states in the traditional BKT model, respectively. [h]

the cognitive state evaluation of the fuzzy KT models, it is determined that the student has decreased cognitive states (0.27→0.20) from time steps 11 to 12 (Case 2-1 in (c)) according to the original slightly declining scores (Case 2-1 in (a)).

Example 2-2: As shown in Fig. 11, the student received slightly rising scores (0.7→0.9) at time steps 33 to 34 (Case 2-2 in (a)). In the traditional KT models, the scores were described as 1→1 (Case 2-2 in (d)). Through the cognitive state evaluation (Case 2-2 in (f)), it is misunderstood that the student has no cognitive change when he achieves consecutive correct answers. However, through the cognitive state evaluation of the fuzzy KT models, it is determined that the student has increased cognitive states (0.69→0.76) from time steps 33 to 34 (Case 2-2 in (c)) according to the original slightly rising scores (Case 2-2 in (a)).

Explainability of the relationship between the performance and cognitive states at a time step: Compared to most existing KT models, including the traditional BKT and deep version models, the fuzzy KT models clearly explain the relationship between the fuzzy cognitive state and performance at a time step.

Specifically, in the fuzzy KT models, the relationships between students' cognitive states and their scores on exercises are described by the following three fuzzy relations: S2FP, FP2OP, and S2OP relations

In the traditional BKT model, the relationships between students' cognitive states and their performance scores on exercises are described as slipping and guessing probabilities, corresponding to the probability of achieving an incorrect answer when knowing the knowledge components, and the probability of achieving a correct answer when not knowing the knowledge components. Therefore, only four cases can be explained by the traditional BKT model, whereas the proposed fuzzy KT models explain more cases.

In the deep version KT models, the relationship between the hidden units and the outputs is unclear. Though the hidden layer plays a vital role in information transmission, there may be several hundred hidden units, which complicates the accurate explanation of every hidden unit at each time step. Therefore, the proposed fuzzy KT models achieve better explainability.

VII. Conclusion

Most existing KT models rely on the bisection representation of student cognitive states and performance. This limits the accuracy of the evaluation of student cognition and the applicability of KT models to continuous score scenarios. By exploiting the ability of the fuzzy theory to handle concepts such as partially correct and partially known, we proposed FBKT to address the aforementioned limitations. Furthermore, T2FBKT was proposed to mitigate model uncertainty by using the type-2 fuzzy method. Finally, extensive experiments demonstrated that the proposed fuzzy KT models achieve satisfactory performance. The fuzzy KT models provide an effective way to extend the application

[h]Bisection performance scores in the traditional KT models (Figure (d)) are obtained by bisection representing the continuous performance scores (Figure (a)), because the traditional KT models cannot address continuous performance scores.

Relationship among the performance scores (Figures (a) and (d)), the probabilities of cognitive states (Figures (b) and (e)), and the cognitive states (Figures (c) and (f)): Performance scores are the observations, from which the probabilities of the cognitive states are calculated from the KT models. Finally, the cognitive states are obtained based on the probabilities of the cognitive states.

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scope of KT models and capture subtler changes in student cognition than existing models. In the future, we will apply the idea of fuzzy KT to deep-learning KT models to improve their interpretability.

Appendix A
Derivation of the Log-Likelihood Function of the Proposed Models

Notation. The model parameter λ and the observed performance sequence $Y = (y_1, y_2, ..., y_t, ..., y_T)$ are given, where y_t denotes the observed performance at time step t. Furthermore, the cognitive state and fuzzy performance sequences are defined as $I = (i_1, i_2, ..., i_t, ..., i_T)$ and $J = (j_1, j_2, ..., j_t, ..., j_T)$, respectively.

Derivation. Given the parameter λ, the probability of the fuzzy cognitive state sequence $I = (i_1, i_2, ..., i_t, ..., i_T)$ satisfies follows:

$$P(I \mid \lambda) = \tilde{\eta}_{i_1}^{(2)}(x_1) \prod \tilde{a}_{i_1 i_2} \prod \cdots \prod \tilde{a}_{i_{T-1} i_T}. \tag{29}$$

Given the HMM parameter λ and the fuzzy cognitive state sequence $I = (i_1, i_2, ..., i_t, ..., i_T)$, the probability of the fuzzy performance sequence $J = (j_1, j_2, ..., j_t, ..., j_T)$ satisfies follows:

$$P(J \mid I, \lambda) = \tilde{b}_{i_1, j_1} \prod \tilde{b}_{i_2, j_2} \prod \cdots \prod \tilde{b}_{i_T, j_T}. \tag{30}$$

Given the HMM parameter λ, the fuzzy cognitive state sequence $I = (i_1, i_2, ..., i_t, ..., i_T)$, and the fuzzy performance sequence $J = (j_1, j_2, ..., j_t, ..., j_T)$, the probability of the observed performance sequence $Y = (y_1, y_2, ..., y_t, ..., y_T)$ satisfies follows:

$$P(Y \mid J, I, \lambda) = \tilde{\eta}_{j_1}^{(1)}(y_1) \prod \tilde{\eta}_{j_1}^{(1)}(y_2) \prod \cdots \prod \tilde{\eta}_{j_T}^{(1)}(y_T). \tag{31}$$

Thus given the HMM parameter λ, the joint probability $P(Y, J, I \mid \lambda)$ satisfies follows:

$$\begin{aligned} P(Y, J, I \mid \lambda) &= P(Y \mid J, I, \lambda) \prod P(J \mid I, \lambda) \prod P(I \mid \lambda) \\ &= \tilde{\eta}_{i_1}^{(2)}(x_1) \prod_{t=1}^{T} \tilde{b}_{i_t, j_t} \prod_{t=1}^{T} \tilde{\eta}_{j_t}^{(1)}(y_t) \prod_{t=1}^{T-1} \tilde{a}_{i_t i_{t+1}}. \end{aligned} \tag{32}$$

Therefore, the log-likelihood function is

$$\begin{aligned} L(\lambda, \hat{\lambda}) &= \coprod_N \coprod_M P(I, J \mid Y, \hat{\lambda}) \log P(Y, I, J \mid \lambda) \\ &= \coprod_N \coprod_M \frac{P(Y, I, J \mid \hat{\lambda})}{P(Y \mid \hat{\lambda})} \log P(Y, I, J \mid \lambda). \end{aligned} \tag{33}$$

$\hat{\lambda}$ can be obtained by maximizing the log-likelihood function.

$$\hat{\lambda} = \arg\max_{\lambda} \coprod_N \coprod_M \frac{P(Y, I, J \mid \hat{\lambda})}{P(Y \mid \hat{\lambda})} \log P(Y, I, J \mid \lambda). \tag{34}$$

Since $P(Y \mid \hat{\lambda})$ is a constant, then

$$\hat{\lambda} = \arg\max_{\lambda} \coprod_N \coprod_M P(Y, I, J \mid \hat{\lambda}) \log P(Y, I, J \mid \lambda). \tag{35}$$

Combining Eqs. (32) and (35), then

$$\begin{aligned} \hat{\lambda} = \arg\max_{\lambda} \coprod_N \coprod_M P(Y, I, J \mid \hat{\lambda}) \log \tilde{\eta}_{i_1}^{(2)}(x_1) \coprod \\ \left[\coprod_{t=1}^{T} \log \tilde{b}_{i_t, j_t} \coprod_{t=1}^{T} \log \tilde{\eta}_{j_t}^{(1)}(y_t) \coprod_{t=1}^{T-1} \log \tilde{a}_{i_t i_{t+1}} \right]. \end{aligned} \tag{36}$$

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